

Shiyuan Piao

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EDUCATION

The Hong Kong University of Science and Technology Doctor of Philosophy in Industrial Engineering and Decision Analytics • Hong Kong Postgraduate Fellowship (HKPFS), supervised by Prof. Fugee Tsung • Selected coursework: data mining, optimization, operations research, decision analytics	Hong Kong, China 2025/09–Present
The Hong Kong University of Science and Technology (Guangzhou) Master of Philosophy in Data Science and Analytics Thrust, Information Hub • Industrial Informatics and Intelligence Institute; Profs. Fugee Tsung and Yuxuan Liang • GPA: 4.12/4.3; Co-founder, UrbanNexus smart-city project funded by HKUST(GZ) (3/73 by students) • Selected coursework: Data Mining; Spatio-temporal Data Mining; Advanced Deep Learning	Guangzhou, China 2023/09–2025/06
South China University of Technology (SCUT, 985 & 211) B.Mgmt. in Management Science, School of Business Administration B.C.S. in Computer Science and Technology, School of Computer Science and Technology • GPA: 91.24/100 (3.88/4.0); Rank: 1/29 (1/106 whole major); Prof. Baozhuang Niu • Minor: Computer Science; Information Engineering • Selected coursework: Calculus, Linear Algebra, Statistics, Operations Research, Deep Learning, Machine Learning	Guangzhou, China 2019/09–2023/06 2020/09–2023/06
McGill University Visiting Student in Machine Learning Group; GPA: 4.0/4.0	2021/07–2021/08

RESEARCH INTERESTS

- Spatio-temporal Data Mining; Uncertainty Quantification; Conformal Prediction; AI in Operations Management; LLM Agents

SELECTED PUBLICATIONS

* Indicates equal contribution; † indicates corresponding author.

Published / Accepted

- CRULP: Reliable RUL Estimation Inspired by Conformal Prediction**
Shiyuan Piao, Ruyi Huang, Fugee Tsung[†]
VENUE: *IEEE Transactions on Instrumentation and Measurement*, 2025 | STATUS: Published | TAGS: SCI; JCR Q1; WOS Highly Cited 2025
Early version nominated for the Best Paper Award at the China National Student Forum on Industrial Engineering, 2024.
- Fourier Geometric Wind Power Forecasting with Numerical Weather Prediction**
Shiyuan Piao, Zehui Fan, Yang Liu, Hong Cheng, Juepeng Zheng, Jie Zhou, and Fugee Tsung[†]
VENUE: *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD 2026)* | STATUS: Accepted | TAGS: CCF A; CORE/ICORE A*
- TA-TFN: Task-Adaptive Frequency Decoupling for Generalizable Battery State Estimation across Chemistries**
Jing Wang, Shiyuan Piao, and Fugee Tsung[†]
VENUE: *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD 2026)* | STATUS: Accepted | TAGS: CCF A; CORE/ICORE A*
- Power Consumption Prediction of Mars Express Orbiter Based on Adaptive Frequency-Domain Pruning-Enhanced Transformer**
Zhuoyi Qian*, Shiyuan Piao*, Huan Wang, Zhen Chen, Ershun Pan, Fugee Tsung[†]
VENUE: *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2026 | STATUS: Published | TAGS: SCI; JCR Q1; CCF B
- Efficient Dual-Observed Leader-Following Consensus Control for Multi-Agent Systems**
Ziming Wang*, Shiyuan Piao*, Yiding Ji, Xin Wang, Fugee Tsung[†]
VENUE: *IEEE International Conference on Automation Science and Engineering (CASE 2025)* | STATUS: Early version accepted
- MCCoder: Streamlining Motion Control with LLM-Assisted Code Generation and Rigorous Verification**
Yin Li, Liangwei Wang, Shiyuan Piao, Boo-Ho Yang, Ziyue Li, Wei Zeng, Fugee Tsung[†]
VENUE: *IEEE International Conference on Automation Science and Engineering (CASE 2025)* | STATUS: Early version accepted
Early version nominated for the Student Best Paper Award in CASE 2025.

Under Review / Revision / In Preparation

- Quantify Uncertainty Beyond Covariate Shift in RUL Estimation by Conformal Prediction**
Shiyuan Piao*, Ying Wang*, Ruyi Huang, Di Wang, Fugee Tsung[†]
VENUE: *IEEE Transactions on Automation Science and Engineering* | STATUS: Under review | TAGS: SCI; JCR Q1
Early version accepted in IEEE International Conference on Automation Science and Engineering (CASE 2025).
- An Automatic Event Prediction Method with Incomplete User Profiles via Graph-Enhanced Temporal Point Processes**
Boyang Li, Lingzheng Zhang, Shiyuan Piao, Yu An, Fugee Tsung, and Xi Zhang[†]
VENUE: *IEEE Transactions on Automation Science and Engineering*, 2026 | STATUS: Conditionally accepted

9. **Adaptive Localized Conformalized Spatio-Temporal Forecasting**

Shiyuan Piao, Yutong Feng, Yuxuan Liang, Fugee Tsung[†]

VENUE: *Advances in Neural Information Processing Systems (NeurIPS 2025)* | STATUS: Under review | TAGS: CCF A; CORE/ICORE A*

Early version invited to talk at the INFORMS Annual Meeting, QSR (Data Mining) Session, 2024.

10. **NeoST: Synthetic Pre-training for Generalizable Spatio-Temporal Foundation Models**

Shiyuan Piao, Yuxuan Liang, Fugee Tsung[†]

VENUE: *Advances in Neural Information Processing Systems (NeurIPS 2026)* | STATUS: Under review | TAGS: CCF A; CORE/ICORE A*

11. **What Deep Learning Tells Us About Credit Rating and Asset Allocation**

Shiyuan Piao, Yan Zeng, Qi Zhou, Fugee Tsung[†]

VENUE: *INFORMS Journal on Computing* | STATUS: Reject and resubmit | TAGS: UTD 24

12. **Hotel Selection Considering Review Sentiment and Multi-Day Route Planning**

Shiyuan Piao, Kaiwen Shen, Shanshi Liu[†]

VENUE: *Tourism Management* | STATUS: Major revision | TAGS: SSCI; ABS 4; JCR Q1

Undergraduate Outstanding Thesis of SCUT and the School of Business Administration.

13. **Reinforcement Learning-Based Optimized Event-Triggered Control for Multi-Agent Systems**

Ziming Wang*, Shiyuan Piao*, Qing-Long Han, Xiaohua Ge, Fugee Tsung[†]

VENUE: *IEEE Transactions on Automatic Control* | STATUS: In preparation | TAGS: SCI; JCR Q1

14. **A Hashtag-Based Semantic Network Group Consensus Method for Online Social Media Group Consensus Reaching**

Kaiwen Shen, Zehua Yi, Shiyuan Piao, Zhang-peng Tian[†]

VENUE: *Information and Management* | STATUS: Under review | TAGS: SCI; ABS 3; JCR Q1

15. **Robust Long-Term Time Series Forecasting Inspired by Uncertainty**

Shiyuan Piao, Yuxuan Liang, Fugee Tsung[†]

VENUE: *INFORMS Journal on Computing* | STATUS: Under review | TAGS: SCI; UTD 24

Early version under preparation for SIGKDD 2025; invited talk at INFORMS Annual Meeting 2025, Atlanta.

RESEARCH EXPERIENCES

Model-Free, Distribution-Free Uncertainty Quantification Beyond Covariate and Distribution Shift

Advance conformal prediction methods for reliable data-driven modeling under dependence, hierarchical classification, distribution shift, and real-time decision settings.

- Develop CP frameworks for multivariate and spatio-temporal data, with applications in weather forecasting and manufacturing.
- Introduce nonconformity measures and uncertainty targets for hierarchical classification and stochastic processes.
- Integrate UQ into foundation models and real-time change-point detection for reliable responses to data shifts.

UrbanNexus: Spatio-temporal Data Mining in Smart Cities

Student-led smart-city project funded by HKUST(GZ), collaborating with industrial and public-sector partners including Huawei, Goldwind, and the City of Tallahassee.

- Propose transformer architectures enhanced by product quantization and retrieval-augmented self-attention for high-dimensional forecasting.
- Apply continuous spatio-temporal point processes to detect and analyze energy disruptions in real time.
- Integrate multi-domain and multi-modal data for air-quality, traffic, and energy forecasting in the Greater Bay Area.

INTERNSHIP

AQUMON

Algorithmic Investment Solutions Group

Shenzhen, China

2022/11–2023/04

- Conducted performance tracking and attribution analysis for Hong Kong and U.S. equity strategies using Barra-style factor exposure analysis.
- Developed event-driven alpha strategy research for A-share markets and summarized strategy performance in weekly trading analysis reports.

Iresearch Consulting Group

Data Analyst

Guangzhou, China

2021/10–2022/01

- Monitored retail promotion prices by scraping product price and sales data from e-commerce and social commerce platforms.
- Collected and analyzed online consumer reviews with NLP-based sentiment and topic analysis to support monthly market intelligence reports.

EXTRACURRICULAR ACTIVITIES

Volunteer Team, Vice President

Youth League Committee, School of Business Administration, SCUT

Guangzhou, China

2020/11–2021/12

- Organized and coordinated monthly volunteer events at the Martyrs' Cemetery, mobilizing over 300 participants.

Volunteer Team, Committee Member

Guangzhou, China

Young Volunteers Association, Affiliate High School of South China Normal University

2016/12–2018/06

- Coordinated bi-weekly volunteer visits to a Down Syndrome kindergarten and provided long-term support for children with special needs.

HONORS & AWARDS

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|---|---------------------------|
| • Best Student Paper Award (Finalist), 2025 Conference of Automation Science and Engineering (CASE) | 2025/08 |
| • Hong Kong Postgraduate Fellowship (HKPFS) by HK RGC (Only 3 recipients in IEDA) | 2025/04 |
| • HKUST RedBird Recruitment PhD Award by HKUST (highest honor at HKUST) | 2025/04 |
| • Nomination for the Red Bird MPhil Final Stage Show by HKUST(GZ) (8/73) | 2025/02 |
| • Best Paper Award (Finalist) in the National Student Forum on Industrial Engineering (6/130) | 2024/12 |
| • Outstanding Thesis of Undergraduate by SCUT and School of Business Administration (8/370+) | 2023/06 |
| • First-class Scholarship by Science and Technology Innovation Competition Fund in SCUT (Top 2%) | 2023/06 |
| • National Scholarship by the Ministry of Education of China (The only recipient of major, Top 1%) | 2022/12 |
| • Nomination of Student of the Year by SCUT and School of Business Administration (6/370+) | 2022/10 |
| • First-class Scholarship by Science and Technology Innovation Competition Fund in SCUT (Top 2%) | 2022/06 |
| • Meritorious Prize in Mathematical Contest in Modeling (Problem B, Top 8%) | 2022/04 |
| • First Prize in Asia-Pacific Mathematical Contest in Modeling (Top 5%) | 2022/01 |
| • First-class Scholarship for undergraduate by SCUT (Top 2%) | 2021/12 |
| • First Prize in CECC Cup Mathematical Contest in Modeling (Top 1%) | 2021/12 |
| • Outstanding student leader by SCUT | 2021/10 |
| • Second Prize in China Undergraduate Mathematical Contest in Modeling | 2021/10 |
| • Second Prize in China Undergraduate Mathematical Contest | 2020/10 |
| • Merit student by SCUT | 2020/10, 2021/10, 2022/10 |

SKILLS & ACTIVITIES

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- **Languages:** Mandarin (native), Korean (native), Cantonese (fluent), English (fluent; TOEFL 107; GRE 155+170)
 - **Computing:** Python / C++ / MATLAB / Linux / Docker / Oracle / MS SQL server / LaTeX / MS Office / Bloomberg / Wind
 - **Data Science:** PyTorch / PyTorch Geometric Temporal / TensorFlow / Scikit-learn / SciPy / Polars / Pandas / NumPy
 - **Personal Interests:** Bicycle touring (cycled around Hainan Province, 960 km), badminton, hiking

ACADEMIC SERVICES

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- **Journal Reviewer:** Production and Operations Management, Manufacturing & Service Operations Management, IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Industrial Informatics, IEEE Transactions on Automation Science and Engineering, IEEE Transactions on Instrumentation and Measurement, International Journal of Production Economics, International Journal of Production Research, Transportation Research Part B/C/E
 - **Conference Reviewer:** ICML 2025, SIGKDD 2025, AAAI 2025, CASE 2025